

Transit Amenity Access Disparities in the Bay Area

Group Number: 13

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Track: B

MTC Team: Yes

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1. Problem & motivation

Updated research question: This project investigates whether Bay Area transit stations provide equitable access to essential amenities for transit-dependent riders. Our central question is: *Do peripheral BART and Caltrain stations provide statistically significantly less access to essential amenities than core, and are these disparities concentrated among stations serving transit-dependent populations?* Our specific sub-questions and findings are discussed in Section 5. *Core* and *peripheral* stations are classified using ridership data from FY2025: *core* stations are high-ridership stops, while *peripheral* stations are lower-ridership stops across the BART and Caltrain networks. *Amenity access* is the count and diversity (amenity type) of essential services: grocery stores, parks, clinics, pharmacies, hospitals, doctors' offices, and childcare, within a half-mile walking buffer. *Transit-dependent* populations are households without a personal vehicle, for whom transit is a primary means of getting around.

Stakeholders and Impact: Stations that serve high proportions of car-free riders but offer few nearby amenities create compounded mobility burdens, with direct implications for transit-oriented development and equitable service planning. The riders most affected are transit-dependent Bay Area residents, particularly low-income households, elderly riders, and people with disabilities, who rely on transit not just for commuting but for reaching daily necessities (*Public Transportation Fact Book - APTA*, 2023). Secondary stakeholders include BART, Caltrain, the MTC, and local planners making land use decisions near station areas.

Unit of Analysis: Our unit of analysis is the 81 individual transit stations. We chose the station level because it directly reflects the walkable environment riders experience in turn, giving us an actionable scale for recommendations around prioritizing nearby development. Each station is matched to its enclosing census tract to retrieve neighborhood demographics.

Measures and Metrics: We supplement this with three derived metrics: a Gini coefficient measuring inequality in amenity distribution across the network; an Unmet Need Index capturing the joint imbalance between car-free household rates and amenity supply, where stations with high need and low supply receive the highest scores; and Shannon entropy capturing amenity type diversity per station. Demographic predictors (median income, percent nonwhite, and percent no-vehicle households) come from the 2024 ACS 5-year estimates at the census tract level, the most recent data available. BART and Caltrain ridership data came from the 2025 fiscal year (July 2024 to June 2025) as Caltrain only provides yearly estimates covering July to June of the following year, which limited us to the FY2025 Annual Ridership Report as the most recent station-level source available. Unlike BART, which reports average weekday exits via faregate counts, Caltrain estimates ridership using a fare media sales model and reports Average Mid-Week Ridership (AMWR: the average of Tuesday through Thursday boardings) at the origin station level. Both metrics serve as comparable metrics for typical weekday station activity.

2. Data & cleaning

Our analysis draws on three datasets. Transit station data comes from the official BART API and Caltrain's GTFS feed via the 511 SF Bay API, giving us 81 stations total with attributes including coordinates, agency, and mode type. Amenity data was collected from OpenStreetMap via the Overpass API across 7 categories (grocery stores, parks, clinics, pharmacies, hospitals, doctors' offices, and childcare facilities) yielding 11,360 point-of-interest records across the 9-county Bay Area. Census demographic data comes from the 2024 ACS 5-year estimates pulled at the census tract level for all 9 Bay Area counties, covering variables on income, race, vehicle ownership, commute mode, housing tenure, and poverty. All three datasets were collected programmatically and saved as CSVs before any processing.

Several cleaning steps were necessary. BART and Caltrain used different naming conventions, so we standardized column names and retained only the fields needed for analysis before merging into a single station dataset. On the amenity side, kindergarten and childcare records came from separate OSM tags but represent the same category, so we consolidated them under a single label. Duplicate stations with identical coordinates were

dropped, keeping the first occurrence. For the census data, ACS encodes missing values as a large negative sentinel number, which we replaced with NaN before computing any derived variables, and all numeric fields were explicitly recast to avoid silent string-typing errors.

The most significant cleaning issue was census tract matching. Our original approach assigned stations to tracts using nearest-centroid logic, which produced implausible demographic patterns for some stations. We rebuilt this entirely using a spatial join against Census TIGER/Line 2024 tract boundary polygons, ensuring each station is matched to the tract it physically falls within rather than the nearest centroid. This matched all 81 stations successfully, with 79 having complete demographic data. The two exceptions (Stanford and SFO) fall within institutional tracts that lack residential household data, which is expected and noted in our missing data documentation.

Key transformations applied to the raw census data include computing percent no-vehicle households, percent nonwhite, poverty rate, and commute mode shares from raw ACS counts. For the station-level analysis dataset, we additionally derived three composite metrics: an Unmet Need Index combining car-free household rates with normalized amenity supply, Shannon entropy scores measuring amenity category diversity per station, and a Gini coefficient capturing inequality in amenity distribution across the full network.

3. Missing data analysis

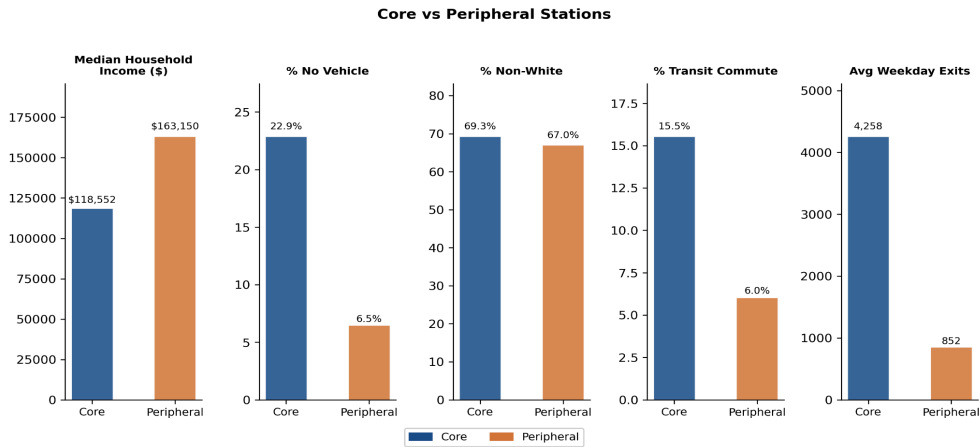
The census data contained missing values, which were denoted as “-666666666”. Any missing values were converted to NaN values. Any rows with null values in the data were dropped. SFO Airport was the only station with missing census data. This is an instance of **Type I Dark Data** (data we know is missing). The tract SFO falls within is an institutional/commercial zone with no residential population, so household-level variables like vehicle ownership and commute mode simply don't exist for it. Excluding SFO is unlikely to bias our results since it doesn't serve a transit-dependent residential population to begin with. Additionally, Stanford Caltrain Station had no available ridership data to be classified as a core or peripheral station. This is also **Type I**. Upon further investigation, we found that the Stanford Station is only operable a few times a year for Stanford University football games and special events. Stanford Caltrain Station was therefore also excluded from analysis.

One potential bias worth noting: OpenStreetMap amenity coverage may be less complete in lower-income or less-mapped neighborhoods, which could cause us to undercount amenities near some high-need stations. This would work against finding disparities, making our estimates conservative rather than inflated.

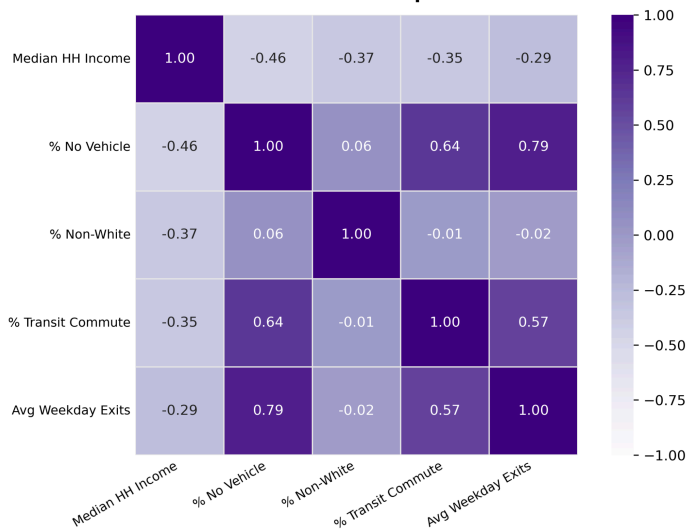
4. Initial results & visualizations

Initial analysis across 79 stations reveals that core stations average 21.9 amenities within a half-mile compared to 9.8 at peripheral stations. The network-wide Gini coefficient of 0.428 indicates moderate inequality in amenity distribution. Coliseum BART scores highest on the Unmet Need Index (0.801), with 29.4% car-free households but only 3 nearby amenities.

To the left is a comparison of demographic averages for core and peripheral stations. Peripheral stations are in census tracts that have a higher median household income, but have significantly lower percentages of households with no vehicles, transit commuters, and ridership. Core and peripheral stations have nearly the same percentage of non-white residents.

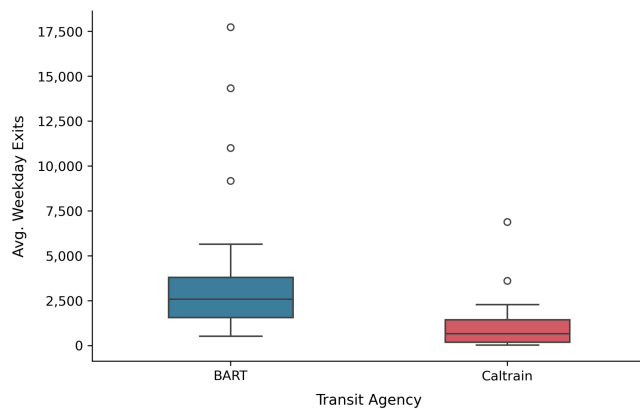


Correlation Matrix of Station-Level Demographics and Ridership



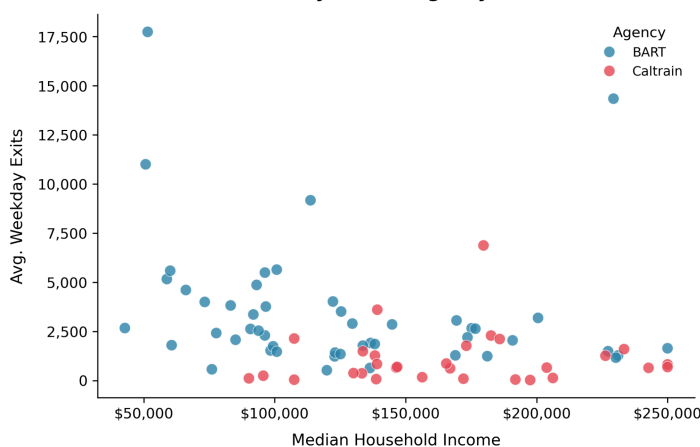
The correlation map on the left shows that there is significant positive ($r^2 > 0.50$) association between percentage of non-vehicle households and average weekday ridership, and percentage of transit commuters and average weekday ridership, which intuitively makes sense. There appears to be a negative association between median household income and average weekday ridership, and percentage of non-white residents and average weekday ridership.

Distribution of Average Weekday Exits by Transit Agency



The boxplot on the left reveals a stark difference in ridership scale between the two agencies. BART stations have a median of roughly 2,500 average weekday exits, with several high-ridership outliers exceeding 10,000. Caltrain stations cluster near 1,000 exits with much less spread, suggesting a more uniformly low-ridership network. This difference in scale gives important context for the core/peripheral classification.

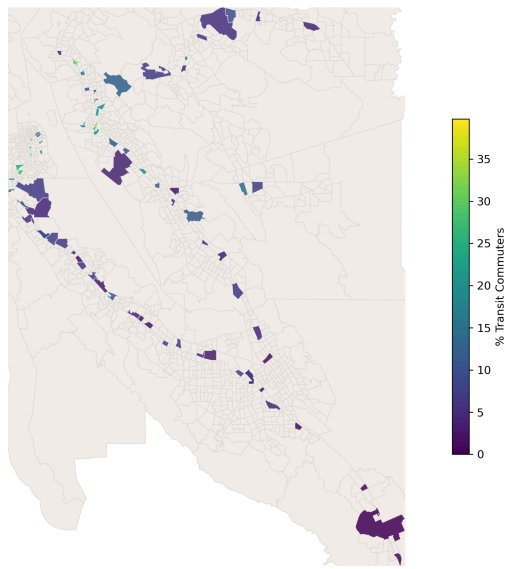
Avg. Weekday Exits vs. Median Household Income by Transit Agency



The scatter plot on the left shows that the median income of Caltrain stations is generally higher than BART, but ridership remains low. Additionally, the points are relatively flat, indicating that there is no relationship between household income and average weekday ridership for Caltrain. For BART, the points are more spread, but show a negative correlation between median household income and ridership.

The map below displays the census tract surrounding each BART and Caltrain station, shaded by the percentage of residents who commute by transit. The darkest purple tracts (concentrated along the Caltrain corridor on the

**Transit Commuter Share by Transit Station
(Bay Area)**



Peninsula and in the outer East Bay) have the lowest transit commuter shares, while the brighter teal and yellow tracts cluster around core urban stations, particularly in San Francisco and Oakland. This geographic pattern suggests that transit commuter share is not evenly distributed across the Bay.

A few limitations are worth flagging here. OSM amenity data tends to be less complete in lower-income and less-mapped neighborhoods, so we may be undercounting amenities near the stations that need them most. If anything, this makes our disparity findings more conservative. Census tract boundaries are also a rough approximation; a station's tract can extend far beyond the half-mile walkshed we actually care about. Finally, the negative correlation between percent nonwhite and ridership is a bit counterintuitive and probably says more about where BART's network is geographically concentrated than anything else.

5. Progress since proposal & plan to completion

Statistical Methods: We originally planned to use chi-square tests and OLS regression, but our data violates the assumptions these require: amenity counts are highly skewed with substantial overdispersion. We therefore use permutation tests (10,000 resamples) for group comparisons and Spearman rank correlations for demographic analyses, as these methods do not require normality or homoscedasticity.

Stations are classified as core or peripheral using FY2025 ridership data through a consensus-based approach, yielding two balanced groups (40 core, 39 peripheral). Because this classification is derived from a continuous measure (average weekday exits) rather than arbitrary geographic labels, it gives us a more grounded cutoff than drawing an arbitrary geographic line. Permutation testing works well here because it doesn't assume anything about how the data is distributed. It just asks whether the differences we see are larger than what you'd get by randomly shuffling station labels.

Spearman correlations are used to assess relationships between amenity counts and demographic variables, including percent no-vehicle households and percent nonwhite, where nonlinearity and outliers are likely. To account for multiple testing across five amenity categories, we apply the Benjamini-Hochberg false discovery rate correction at $\alpha = 0.05$. Effect sizes are reported using Glass's Δ rather than Cohen's d due to unequal group variances; Glass's Δ uses the peripheral group's standard deviation as the reference, which makes more sense here since our two groups have pretty different spreads.

Enhanced Amenity Metrics: Based on feedback, we developed three additional metrics (described in Section 1). Using these updated metrics we found Coliseum BART ranks highest (index = 0.250), with 29.4% car-free households but only 3 nearby amenities. Shannon entropy scores measure amenity diversity per station; core stations score slightly higher (0.923 vs. 0.759), though this difference is not statistically significant ($p = 0.2047$).

Core-Peripheral Classification: Stations are classified using average weekday exit ridership from FY2025 data, replacing an earlier judgment-based approach. A consensus-based method applied to the ridership distribution yields two balanced groups (40 core, 39 peripheral). This approach is more defensible than our original classification, which required judgment calls on borderline stations such as Walnut Creek and Concord. Stanford and San Francisco International Airport are excluded as outliers, as their ridership patterns do not reflect typical commuter or transit-dependent use.

Research Question Refinement: We restructured our analysis around four sub-questions.

- SQ1 (Quantity): Is the raw count of amenities within 0.5 miles significantly lower at peripheral stations than core stations?
- SQ2 (Diversity): Do peripheral stations have access to fewer categories of amenities (e.g., a station near a park but no grocery store)?
- SQ3 (Equity): Are stations serving higher proportions of non-white residents, low-income households, or zero-vehicle households more likely to be amenity-poor?
- SQ4 (Agency difference): Does the BART vs. Caltrain distinction predicts amenity access independently of core/peripheral classification?

Scope Adjustments: Our data scope has remained largely consistent with the proposal: 81 stations, 11,360 amenities from OpenStreetMap, and census demographics across 9 Bay Area counties. We expanded amenity categories, adding hospitals, doctors' offices, childcare and kindergarten facilities, and convenience stores. We also excluded 2 stations from our overall analysis due to non-residential tract classifications, yielding a valid sample of 79 stations.

Completed Work and Remaining Tasks: Data collection, cleaning, and spatial matching are complete, along with most of our statistical analysis framework, including the Gini coefficient, Unmet Need Index, Shannon entropy calculations, and correlation analysis. We also implemented a data-driven core/peripheral reclassification using ridership. Remaining tasks include addressing subquestions three and four and developing visualizations to support our statistical findings for the final report. We will create two interactive maps: one visualizing individual amenities within a half-mile radius of each station, and another showing all stations within a two- or three-pane interface consisting of a zoomable, clickable map, a linked detailed station view, and an optional comparison panel displaying the selected station relative to others.

Timeline: The midterm report is due March 31, following which we will finalize visualizations.. Independent work on reclassification and the interactive map will continue through April 14–21 while our mentor is unavailable. Check-ins with Elliot (our project mentor) are scheduled for April 29 and May 6, with the final project due May 10 and presentations the week of May 12. Team responsibilities are divided across statistical justification, visualization and classification, and methodology write-up, with Sunday meetings for integration and GitHub upkeep.

6. References

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Github Repo: https://github.com/sonyak28/City_Planning_255